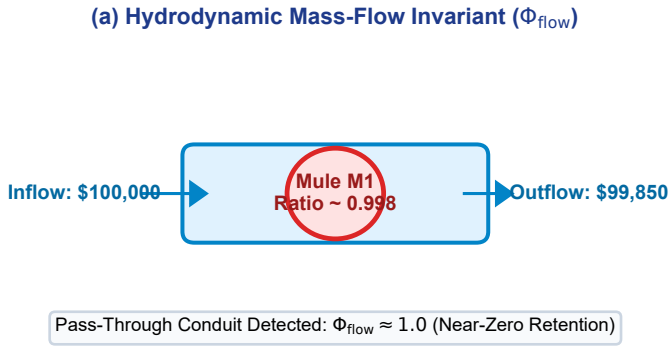
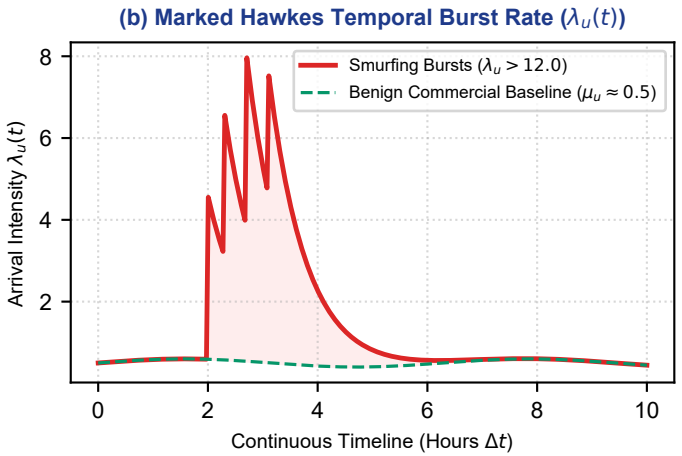
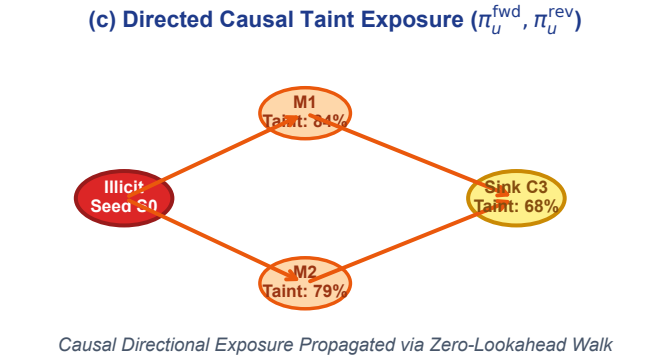
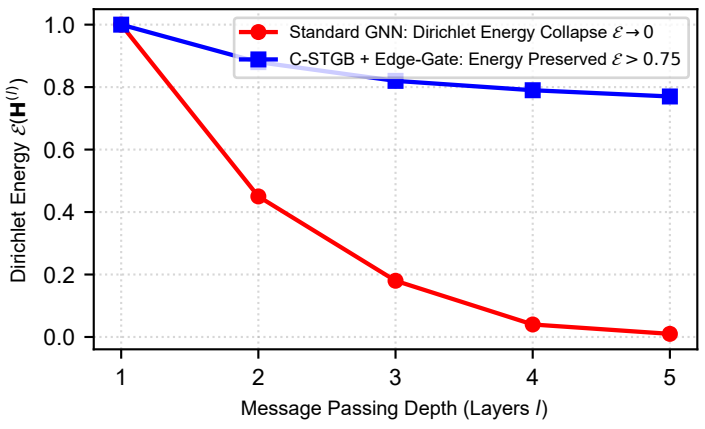

(a) Hydrodynamic Mass-Flow Invariant (Φ_{flow})
Inflow: $100,000
Mule M1
Ratio ~ 0.998
Outflow: $99,850
Pass-Through Conduit Detected: $\Phi_{flow} \approx 1.0$ (Near-Zero Retention)


(b) Marked Hawkes Temporal Burst Rate ($\lambda_u(t)$)
Smurfing Bursts ($\lambda_u > 12.0$)
Benign Commercial Baseline ($\mu_u \approx 0.5$)
Arrival Intensity $\lambda_u(t)$
Continuous Timeline (Hours Δt)


(c) Directed Causal Taint Exposure (π_u^{fwd}, π_u^{rev})
M1
Taint: 84%
Illicit
Seed 90
Sink C3
Taint: 68%
M2
Taint: 79%
Causal Directional Exposure Propagated via Zero-Lookahead Walk


(d) Spectral Dirichlet Energy Preservation ($\mathcal{E}(\mathbf{H})$)
Standard GNN: Dirichlet Energy Collapse $\mathcal{E} \to 0$
C-STGB + Edge-Gate: Energy Preserved $\mathcal{E} > 0.75$
Dirichlet Energy $\mathcal{E}(\mathbf{H}^{(l)})$
Message Passing Depth (Layers l)